GOVERNED MODEL REFERENCE

AppSheet Data Model Reference

Money database | Loan Manager ผู้จัดการสินเชื่อ | Loan Business Insights

Foundation reference for future development, impact analysis, and governed reporting

Control	Value
Document status	Baseline - live configuration inventory
Configuration inspected	29 August 2026 (Asia/Singapore)
Database	Money database - 10 tables
OLTP app	Loan Manager ผู้จัดการสินเชื่อ - 6 loaded tables, 2 slices
OLAP app	Loan Business Insights - 11 loaded tables, 11 slices
External helper	Calendar Help Table / Sheet1 - loaded only in OLAP

Authority The AppSheet Database and live app editors remain authoritative. Update this document in the same change set whenever a table, Ref, physical column, virtual column, formula, or slice changes.

1. Scope and maintenance contract

This document records the current structure of the shared operational database, the OLTP Operations app, and the OLAP Reporting app. It is a model inventory and dependency reference; it does not expose production records, borrower documents, bank details, or credentials.

1.1 Source and confidence

Live sources inspected: AppSheet My Apps, Money database, OLTP Data > Columns/Slices, and OLAP Data > Columns/Slices

Evidence basis: Editor-reported table sources, column types and formulas, slice definitions, Ref targets, and Is a part of? settings

Testing claim: No live workflow or production-data test was performed; this was a read-only configuration review

Backup status: Not applicable - no AppSheet or dashboard configuration was changed

1.2 Required update procedure

Step 1 - Inspect: Read the affected database table and both app column/slice definitions before editing this file.

Step 2 - Back up: For an implementation change, record affected AppSheet version numbers before development, as required by AGENTS.md.

Step 3 - Update: Change the component catalog, relationship table, slice catalog, and affected table dictionary in one document revision.

Step 4 - Verify: Reconcile physical columns to the source schema, Ref targets to the Type Details panel, and VC formulas to each app.

Step 5 - Record: Set the baseline date/status and summarize changed dependencies in the change notes.

Change discipline Do not infer payment, delinquency, settlement, or default state from row existence. Preserve auditable posted amounts, reversals/adjustments, due dates, grace rules, and an explicit as-of date.

2. Architecture overview

Component	Named asset	Current scope	Ownership / purpose
Money database	Operational system of record	10 AppSheet Database tables	Physical keys, stored fields, and referential integrity
OLTP Operations	Loan Manager ผู้จัดการสินเชื่อ	6 loaded tables; 2 slices	Data entry, validation, collection workflow, and lightweight operational indicators
OLAP Reporting	Loan Business Insights	11 loaded tables; 11 slices	Historical analytics, partner analysis, loan assessment, and curated reporting feeds
External helper	Calendar Help Table / Sheet1	1 Google Sheets table	Date-spine calculations for the OLAP loan-assessment model

The OLTP and OLAP apps share the same Money database. Application-specific variance is primarily expressed through which tables are loaded, which VCs exist, and which slices expose curated row/column populations.

3. Table catalog

Table	Source	Loaded in	Physical	OLTP VCs	OLAP VCs	Purpose
Borrowers	Money database	OLTP + OLAP	6	24	50	Borrower master records and borrower-level operational or portfolio indicators.
Calendar	Calendar Help Table / Sheet1	OLAP only	2	0	22	Google Sheets-backed date spine used by OLAP loan-assessment projections; not part of Money database.
Cash Pool Contributions	Money database	OLAP only	7	0	2	Partner capital-contribution movements that build the lendable cash pool.
Charges	Money database	OLTP + OLAP	7	15	18	Scheduled amounts due and collection state at charge level, including principal and interest allocation.
Daily Analytics	Money database	OLAP only	17	0	6	Persisted daily portfolio snapshot fact table for historical reporting and governed feeds.
Loan Assessment	Money database	OLAP only	6	0	16	Input/control record for evaluating a proposed loan against projected coverage and return.
Loans	Money database	OLTP + OLAP	19	19	25	Loan contracts, commercial terms, lifecycle state, balances, and loan-level risk metrics.
Partners	Money database	OLTP + OLAP	5	0	12	Partner master records used for capital participation and settlement reporting.
Repayments	Money database	OLTP + OLAP	9	4	13	Posted payment transactions and their allocation to loans and charges.
Settlements	Money database	OLAP only	8	0	0	Auditable partner settlement records and settlement status.
Statistics	Money database	OLTP + OLAP	3	23	36	Small KPI/control table used to expose app-level operational and reporting aggregates.

Physical count includes the AppSheet/Sheets row ordinal where exposed. Row ID is the stable key for AppSheet Database tables; Calendar uses Date as its key.

4. Relationship model

Child table	Ref column	Parent table	Cardinality	Is a part of?
Calendar	Assessment Borrower	Borrowers	Borrowers 1 -> many Calendar	Independent child lifecycle
Cash Pool Contributions	Ref Partner	Partners	Partners 1 -> many Cash Pool Contributions	Independent child lifecycle
Charges	Ref Loans	Loans	Loans 1 -> many Charges	Child rows are marked part-of parent
Loan Assessment	Ref Borrower	Borrowers	Borrowers 1 -> many Loan Assessment	Independent child lifecycle
Loans	Ref Borrowers	Borrowers	Borrowers 1 -> many Loans	Child rows are marked part-of parent
Repayments	Ref Loans	Loans	Loans 1 -> many Repayments	Independent child lifecycle
Repayments	Ref Charges	Charges	Charges 1 -> many Repayments	Child rows are marked part-of parent
Settlements	Ref Partner	Partners	Partners 1 -> many Settlements	Independent child lifecycle

Cascade caution Is a part of? is enabled for Loans.Ref Borrowers, Charges.Ref Loans, and Repayments.Ref Charges. AppSheet describes these rows as line items and may delete them with the referenced parent; financially material corrections should still use an auditable void, reversal, or adjustment process.

Reverse relationships such as Related Loans, Related Repayments, Related Cash Pool Contributions, and Related Settlements are VCs generated or configured from these direct Ref links. Their exact formulas appear in the table dictionaries.

5. Slice catalog

App	Slice	Source	Row filter	Mode	Cols	Purpose
OLTP	Todays Collection Board	Borrowers	ISNOTBLANK([Daily Collection Status])	Read-only	10	Read-only borrower workboard restricted to borrowers with a current collection status.
OLTP	Payment Collection	Charges	OR([Payment Status] = "รอชำระ", [Payment Status] = "ชำระบางส่วน", AND([Payment Status] = "ชำระแล้ว", [Payment Date] = TODAY()))	Updates, Adds, Deletes	22	Operational charge queue containing pending or partially paid items plus items completed today.
OLAP	Borrowers with Outstanding Loans	Borrowers	[Total Outstanding Principal] > 0	Read-only	56	Read-only OLAP subset of borrowers whose outstanding principal is greater than zero.
OLAP	Profit Contribution 30D	Borrowers	[Profit Last 30 Days] > 0	Read-only	56	Read-only OLAP subset of borrowers contributing positive interest profit in the last 30 days.
OLAP	Reporting Borrowers	Borrowers	TRUE / all rows	Read-only	40	Curated, read-only borrower feed for downstream reporting and dashboard reconciliation.
OLAP	Reporting Charges	Charges	TRUE / all rows	Read-only	20	Curated, read-only charge feed with due, paid, remaining, lateness, and borrower fields.
OLAP	Daily Analytics 7 Days	Daily Analytics	[Snapshot Date] >= NOW() - 7	Read-only	23	Rolling seven-day daily-analytics window for recent trend analysis.
OLAP	Daily Analytics 30 Days	Daily Analytics	[Snapshot Date] >= NOW() - 30	Read-only	23	Rolling 30-day daily-analytics window for monthly and near-term trend analysis.
OLAP	Daily Analytics 90 Days	Daily Analytics	[Snapshot Date] >= NOW() - 90	Read-only	23	Rolling 90-day daily-analytics window for medium-term portfolio trends.
OLAP	Reporting Daily Analytics	Daily Analytics	TRUE / all rows	Read-only	13	Curated, read-only daily snapshot feed for governed downstream reporting.
OLAP	Reporting Loans	Loans	TRUE / all rows	Read-only	30	Curated, read-only loan feed containing contract, balance, rate, age, and default fields.
OLAP	Reporting Repayments	Repayments	TRUE / all rows	Read-only	8	Curated, read-only posted-payment feed with loan and charge references.
OLAP	Reporting Statistics	Statistics	TRUE / all rows	Read-only	17	Curated, read-only KPI feed for governed dashboard totals.

5.1 Slice column membership

Todays Collection Board (OLTP)

Source / filter: Borrowers / ISNOTBLANK([Daily Collection Status])

Update mode: Read-only

Included columns: _RowNumber, Row ID, Borrower Name, Description, Daily Collection Status - Order, Todays Amount Due, Todays Amount Collected, Todays Amount Remaining, Todays Collection Charges, Borrower Name - Collection

Payment Collection (OLTP)

Source / filter: Charges / OR([Payment Status] = "รอชำระ", [Payment Status] = "ชำระบางส่วน", AND([Payment Status] = "ชำระแล้ว", [Payment Date] = TODAY()))

Update mode: Updates, Adds, Deletes

Included columns: _RowNumber, Row ID, Ref Loans, Charge Date, Interest Due, Amount Due, Borrower Name, Related Repayments, Payment Status, Payment Date, Borrower Name TE, Payment Status TE, Principal Paid, Interest Paid, Total Paid, Principal Remaining, Interest Remaining, Amount Remaining, Notes, Payment Count, Days Outstanding, Principal Due

Borrowers with Outstanding Loans (OLAP)

Source / filter: Borrowers / [Total Outstanding Principal] > 0

Update mode: Read-only

Included columns: _RowNumber, Row ID, Borrower Name, Telephone, Description, Creation Date, Related Loans, Total Interest Earned, Total Amount Loaned, Total Number of Loans, Total Outstanding Principal, Header - Active Loans, Borrower Name TE, Return on Active Principal, Profit Last 30 Days, Projected Interest Coverage Date, Active Loan Interest Earned, Active Daily Interest, Related Loan Assessments, Related Calendars, Closed Loan Interest Earned, Closed Loan Principle, Has Closed Loan, Return on Closed Loans, Header - Closed Loans, Has Active Loan, 30D Gross Principal Days, 30D Principal Reduction Days, 30D Average Outstanding Principal, 30D Return on Average Principal, Daily Collection Status, Todays Collection Charges, Todays Amount Due, Todays Amount Collected, Todays Amount Remaining, Portfolio Exposure %, Active Loan Count, Oldest Active Loan Age, Overdue Amount, Overdue Charge Count, Historical Charge Count, On Time Charge Count, Lifetime On Time Rate, Recent 30D Charge Count, Recent 30D On Time Count, Recent 30D On Time Rate, Lifetime Late Payment Count, Recent 30D Late Payment Count, Maximum Days Late, Last Late Payment Date, Largest Successful Loan, Average Successful Loan Size, Default Count, Lifetime Default Loss, Portfolio Outstanding Principal, Portfolio Share Join Key

Profit Contribution 30D (OLAP)

Source / filter: Borrowers / [Profit Last 30 Days] > 0

Update mode: Read-only

Included columns: _RowNumber, Row ID, Borrower Name, Telephone, Description, Creation Date, Related Loans, Total Interest Earned, Total Amount Loaned, Total Number of Loans, Total Outstanding Principal, Header - Active Loans, Borrower Name TE, Return on Active Principal, Profit Last 30 Days, Projected Interest Coverage Date, Active Loan Interest Earned, Active Daily Interest, Related Loan Assessments, Related Calendars, Closed Loan Interest Earned, Closed Loan Principle, Has Closed Loan, Return on Closed Loans, Header - Closed Loans, Has Active Loan, 30D Gross Principal Days, 30D Principal Reduction Days, 30D Average Outstanding Principal, 30D Return on Average Principal, Daily Collection Status, Todays Collection Charges, Todays Amount Due, Todays Amount Collected, Todays Amount Remaining, Portfolio Exposure %, Active Loan Count, Oldest Active Loan Age, Overdue Amount, Overdue Charge Count, Historical Charge Count, On Time Charge Count, Lifetime On Time Rate, Recent 30D Charge Count, Recent 30D On Time Count, Recent 30D On Time Rate, Lifetime Late

Payment Count, Recent 30D Late Payment Count, Maximum Days Late, Last Late Payment Date, Largest Successful Loan, Average Successful Loan Size, Default Count, Lifetime Default Loss, Portfolio Outstanding Principal, Portfolio Share Join Key

Reporting Borrowers (OLAP)

Source / filter: Borrowers / TRUE

Update mode: Read-only

Included columns: _RowNumber, Row ID, Borrower Name, Borrower Name TE, Description, Total Interest Earned, Total Amount Loaned, Total Number of Loans, Total Outstanding Principal, Daily Collection Status, Todays Amount Due, Todays Amount Collected, Todays Amount Remaining, 30D Return on Average Principal, Return on Active Principal, 30D Average Outstanding Principal, Profit Last 30 Days, Has Active Loan, Has Closed Loan, Overdue Amount, Overdue Charge Count, Oldest Active Loan Age, Active Loan Count, Portfolio Exposure %, Historical Charge Count, On Time Charge Count, Lifetime On Time Rate, Recent 30D Charge Count, Recent 30D On Time Count, Recent 30D On Time Rate, Lifetime Late Payment Count, Recent 30D Late Payment Count, Maximum Days Late, Last Late Payment Date, Largest Successful Loan, Average Successful Loan Size, Default Count, Lifetime Default Loss, Portfolio Outstanding Principal, Portfolio Share Join Key

Reporting Charges (OLAP)

Source / filter: Charges / TRUE

Update mode: Read-only

Included columns: _RowNumber, Row ID, Ref Loans, Charge Date, Interest Due, Principal Due, Amount Due, Payment Date, Payment Status TE, Principal Paid, Interest Paid, Total Paid, Principal Remaining, Interest Remaining, Amount Remaining, Payment Status, Days Late, Historical On Time, Historical Late, Borrower Name

Daily Analytics 7 Days (OLAP)

Source / filter: Daily Analytics / [Snapshot Date] >= NOW() - 7

Update mode: Read-only

Included columns: _RowNumber, Row ID, Snapshot Date, Generated At, Model Version, Principal Issued, Loans Issued, Principal Returned, Interest Received, Repayments Count, Outstanding Principal EOD, Active Loans EOD, Active Borrowers EOD, Pending Charges EOD, Total Cash Pool EOD, Available Cash EOD, Unsettled Profit EOD, Week, Month, Daily Capital Employed, Daily Capital Yield, Portfolio Share Join Key, Portfolio Slice

Daily Analytics 30 Days (OLAP)

Source / filter: Daily Analytics / [Snapshot Date] >= NOW() - 30

Update mode: Read-only

Included columns: _RowNumber, Row ID, Snapshot Date, Generated At, Model Version, Principal Issued, Loans Issued, Principal Returned, Interest Received, Repayments Count, Outstanding Principal EOD, Active Loans EOD, Active Borrowers EOD, Pending Charges EOD, Total Cash Pool EOD, Available Cash EOD, Unsettled Profit EOD, Week, Month, Daily Capital Employed, Daily Capital Yield, Portfolio Share Join Key, Portfolio Slice

Daily Analytics 90 Days (OLAP)

Source / filter: Daily Analytics / [Snapshot Date] >= NOW() - 90

Update mode: Read-only

Included columns: _RowNumber, Row ID, Snapshot Date, Generated At, Model Version, Principal Issued, Loans Issued, Principal Returned, Interest Received, Repayments Count, Outstanding Principal EOD, Active Loans EOD, Active Borrowers EOD, Pending Charges EOD, Total Cash Pool EOD, Available Cash EOD, Unsettled Profit EOD, Week, Month, Daily Capital Employed, Daily Capital Yield, Portfolio Share Join Key, Portfolio Slice

Reporting Daily Analytics (OLAP)

Source / filter: Daily Analytics / TRUE

Update mode: Read-only

Included columns: _RowNumber, Row ID, Snapshot Date, Principal Issued, Principal Returned, Interest Received, Outstanding Principal EOD, Active Loans EOD, Active Borrowers EOD, Pending Charges EOD, Daily Capital Yield, Portfolio Share Join Key, Portfolio Slice

Reporting Loans (OLAP)

Source / filter: Loans / TRUE

Update mode: Read-only

Included columns: _RowNumber, Row ID, Loan Date, Close Date, Principal Amount, Due Date, Ref Borrowers, Total Interest Received, Total Principal Received, Outstanding Principal, Total Amount Received, Loan Status TE, Loan Age Bucket, Loan Type TE, Loan Status, Loan Type, Current Daily Interest, Fixed Interest, Transfer Fee, Interest Payment Interval, Principal Recovered by Interest %, Expected Interest This Month, Effective Daily Interest Rate, Default Loss Amount, Defaulted Flag, Borrower Name, Borrower Name TE, Loan Duration, Defaulted, Maximum Days Late

Reporting Repayments (OLAP)

Source / filter: Repayments / TRUE

Update mode: Read-only

Included columns: _RowNumber, Row ID, Payment Date, Amount Received, Principal Paid, Interest Paid, Ref Loans, Ref Charges

Reporting Statistics (OLAP)

Source / filter: Statistics / TRUE

Update mode: Read-only

Included columns: _RowNumber, Row ID, Outstanding Principal, Active Loans, Realized Profit MTD, Projected Monthly Profit, Peak Cash Pool Date 30D, Expected Daily Interest, Total Cash Pool, Total Unsettled Profit, Expected Loan Account Balance, Top 3 Borrower Concentration, Amount Due Today, Amount Collected Today, Amount Outstanding Today, Amount Collection Rate Today, Total Outstanding Principal

6. Detailed table dictionaries

Each dictionary is the union of the two app definitions. Availability is column-specific. For a shared VC whose formula differs between apps, the Definition cell records both expressions. Physical fields are authoritative source columns; VCs are application calculations.

6.1 Borrowers

Source: Money database / Borrowers (AppSheet Database)

Loaded in: OLTP + OLAP

Purpose: Borrower master records and borrower-level operational or portfolio indicators.

Column	Kind / type	Availability	Flags	Purpose	Definition / expression
_RowNumber	Physical / Number	OLTP + OLAP	-	System-generated source-row ordinal; not a stable business key.	Stored source column
Row ID	Physical / Text	OLTP + OLAP	Key	Immutable row key generated by the source database.	Stored source column; initial value: UNIQUEID("PackedUUID")
Borrower Name	Physical / Name	OLTP + OLAP	-	Stores the authoritative source value for Borrower Name.	Stored source column
Telephone	Physical / Phone	OLTP + OLAP	-	Stores the authoritative source value for Telephone.	Stored source column
Description	Physical / LongText	OLTP + OLAP	-	Stores the authoritative source value for Description.	Stored source column
Creation Date	Physical / Date	OLTP + OLAP	-	Stores the source date for Creation Date.	Stored source column; initial value: TODAY()
Related Loans	VC / List	OLTP + OLAP	-	Reverse-reference list of child rows that point to this record.	REF_ROWS("Loans", "Ref Borrowers")
Total Interest Earned	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	SUM([Related Loans][Total Interest Received])
Total Amount Loaned	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	SUM([Related Loans][Principal Amount])
Total Number of Loans	VC / Number	OLTP + OLAP	-	Derived count used for workload, portfolio, or risk analysis.	COUNT([Related Loans])
Total Outstanding Principal	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	SUM([Related Loans][Outstanding Principal])
Header - Active Loans	VC / Text	OLTP + OLAP	-	Presentation-oriented label or bilingual display value.	"—— เงินกู้ที่ยังไม่ปิด / Active Loans ——"
Borrower Name TE	VC / Text	OLTP + OLAP	Label	Presentation-oriented label or bilingual display value.	CONCATENATE([Borrower Name], " - ", [Description])
Return on Active Principal	VC / Percent	OLTP + OLAP	-	Derived ratio or performance measure calculated from governed source values.	IF(SUM(SELECT(Loans[Outstanding Principal], AND([Ref Borrowers] = [_THISROW].[Row ID], [Loan Status] = "ยังไม่ปิดยอด"))) > 0, SUM(SELECT(Loans[Total Interest Received], AND([Ref Borrowers] = [_THISROW].[Row ID], [Loan Status] = "ยังไม่ปิดยอด"))) / SUM(SELECT(Loans[Outstanding Principal], AND([Ref Borrowers] = [_THISROW].[Row ID], [Loan Status] = "ยังไม่ปิดยอด"))), 0)
Profit Last 30 Days	VC / Price	OLAP only	-	Derived date, time-window, age, or delinquency measure.	SUM(SELECT(Repayments[Interest Paid], AND([Ref Loans].[Ref Borrowers] = [_THISROW].[Row ID], [Payment Date] >= TODAY() - 29, [Payment Date] <= TODAY(), [Interest Paid] > 0)))
Active Loan Interest Earned	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	SUM(SELECT(Loans[Total Interest Received], AND([Ref Borrowers] = [_THISROW].[Row ID], [Loan Status] = "ยังไม่ปิดยอด")))
Active Daily Interest	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	SUM(SELECT(Loans[Current Daily Interest], AND([Ref Borrowers] = [_THISROW].[Row ID], [Loan Status] = "ยังไม่ปิดยอด", [Loan Type] = "ดอกเบี้ยรายวัน")))

Column	Kind / type	Availability	Flags	Purpose	Definition / expression
Projected Interest Coverage Date	VC / Date	OLTP + OLAP	-	Derived ratio or performance measure calculated from governed source values.	IFS([Total Outstanding Principal] <= 0, "", [Active Loan Interest Earned] >= [Total Outstanding Principal], TODAY(), [Active Daily Interest] > 0, TODAY() + CEILING(([Total Outstanding Principal] - [Active Loan Interest Earned]) / [Active Daily Interest]), TRUE, "")
Related Loan Assessments	VC / List	OLAP only	-	Reverse-reference list of child rows that point to this record.	REF_ROWS("Loan Assessment", "Ref Borrower")
Related Calendars	VC / List	OLAP only	-	Reverse-reference list of child rows that point to this record.	REF_ROWS("Calendar", "Assessment Borrower")
Closed Loan Interest Earned	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	[Total Interest Earned] - [Active Loan Interest Earned]
Closed Loan Principle	VC / Price	OLTP + OLAP	-	Derived AppSheet value used by the app workflow, presentation, or reporting model.	[Total Amount Loaned] - [Total Outstanding Principal]
Has Closed Loan	VC / Yes/No	OLTP + OLAP	-	Derived business-state or risk indicator.	COUNT(SELECT([Related Loans][Row ID], [Loan Status] = "ปิดยอดแล้ว")) > 0
Return on Closed Loans	VC / Percent	OLTP + OLAP	-	Derived ratio or performance measure calculated from governed source values.	IF(AND([Has Closed Loan], [Closed Loan Principle] > 0), [Closed Loan Interest Earned] / [Closed Loan Principle], 0)
Header - Closed Loans	VC / Text	OLTP + OLAP	-	Presentation-oriented label or bilingual display value.	"—— เงินกู้ที่ปิดแล้ว / Closed Loans ——"
Has Active Loan	VC / Yes/No	OLTP + OLAP	-	Derived business-state or risk indicator.	COUNT(SELECT([Related Loans][Row ID], [Loan Status] = "ยังไม่ปิดยอด")) > 0
30D Gross Principal Days	VC / Decimal	OLAP only	-	Derived date, time-window, age, or delinquency measure.	SUM(SELECT(Loans[30D Principal Days], [Ref Borrowers] = [_THISROW].[Row ID]))
30D Principal Reduction Days	VC / Decimal	OLAP only	-	Derived date, time-window, age, or delinquency measure.	SUM(SELECT(Repayments[30D Principal Reduction Days], [Ref Loans].[Ref Borrowers] = [_THISROW].[Row ID]))
30D Average Outstanding Principal	VC / Decimal	OLAP only	-	Derived date, time-window, age, or delinquency measure.	([30D Gross Principal Days] - [30D Principal Reduction Days]) / 30
30D Return on Average Principal	VC / Percent	OLAP only	-	Derived ratio or performance measure calculated from governed source values.	IF([30D Average Outstanding Principal] > 0, [Profit Last 30 Days] / [30D Average Outstanding Principal], 0)
Daily Collection Status	VC / Text	OLTP + OLAP	-	Derived business-state or risk indicator.	IFS(COUNT([Todays Collection Charges]) = 0, "", COUNT(SELECT(Charges[Row ID], AND(IN([Row ID], [_THISROW].[Todays Collection Charges]), [Charge Date] < TODAY(), [Payment Status] <> "ชำระแล้ว"))) > 0, "Overdue", COUNT(SELECT(Charges[Row ID], AND(IN([Row ID], [_THISROW].[Todays Collection Charges]), [Payment Status] <> "ชำระแล้ว"))) = 0, "Fully Paid", OR(COUNT(SELECT(Charges[Row ID], AND(IN([Row ID], [_THISROW].[Todays Collection Charges]), [Payment Status] = "ชำระบางส่วน"))) > 0, AND(COUNT(SELECT(Charges[Row ID], AND(IN([Row ID], [_THISROW].[Todays Collection Charges]), [Payment Status] = "ชำระแล้ว"))) > 0, COUNT(SELECT(Charges[Row ID], AND(IN([Row ID], [_THISROW].[Todays Collection Charges]), [Payment Status] <> "ชำระแล้ว"))) > 0)), "Partially Paid", TRUE, "Not Paid")
Todays Collection Charges	VC / List	OLTP + OLAP	-	Derived date, time-window, age, or delinquency measure.	SELECT(Charges[Row ID], AND([Ref Loans].[Ref Borrowers] = [_THISROW].[Row ID], OR(AND([Payment Status] <> "ชำระแล้ว", [Charge Date] <= TODAY()), AND([Payment Status] = "ชำระแล้ว", [Payment Date] = TODAY()))))
Todays Amount Due	VC / Price	OLTP + OLAP	-	Derived date, time-window, age, or delinquency measure.	SUM([Todays Collection Charges][Amount Due])

Column	Kind / type	Availability	Flags	Purpose	Definition / expression
Todays Amount Collected	VC / Price	OLTP + OLAP	-	Derived date, time-window, age, or delinquency measure.	SUM([Todays Collection Charges][Total Paid])
Todays Amount Remaining	VC / Price	OLTP + OLAP	-	Derived date, time-window, age, or delinquency measure.	SUM([Todays Collection Charges][Amount Remaining])
Portfolio Exposure %	VC / Percent	OLAP only	-	Derived ratio or performance measure calculated from governed source values.	IF(SUM(Borrowers[Total Outstanding Principal]) = 0, 0, [Total Outstanding Principal] / SUM(Borrowers[Total Outstanding Principal]))
Active Loan Count	VC / Number	OLAP only	-	Derived count used for workload, portfolio, or risk analysis.	COUNT(SELECT(Loans[Row ID], AND([Ref Borrowers] = [_THISROW].[Row ID], [Loan Status] = "ยังไม่ปิดยอด")))
Oldest Active Loan Age	VC / Number	OLAP only	-	Derived date, time-window, age, or delinquency measure.	MAX(SELECT(Loans[Loan Age Days], AND([Ref Borrowers] = [_THISROW].[Row ID], [Loan Status] = "ยังไม่ปิดยอด")))
Overdue Amount	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	SUM(SELECT(Charges[Amount Remaining], AND([Ref Loans].[Ref Borrowers] = [_THISROW].[Row ID], [Charge Date] < TODAY(), [Payment Status] <> "ชำระแล้ว")))
Overdue Charge Count	VC / Number	OLAP only	-	Derived count used for workload, portfolio, or risk analysis.	COUNT(SELECT(Charges[Row ID], AND([Ref Loans].[Ref Borrowers] = [_THISROW].[Row ID], [Charge Date] < TODAY(), [Payment Status] <> "ชำระแล้ว")))
Historical Charge Count	VC / Number	OLAP only	-	Derived count used for workload, portfolio, or risk analysis.	COUNT(SELECT(Charges[Row ID], AND([Ref Loans].[Ref Borrowers] = [_THISROW].[Row ID], [Charge Date] < TODAY())))
On Time Charge Count	VC / Number	OLAP only	-	Derived count used for workload, portfolio, or risk analysis.	COUNT(SELECT(Charges[Row ID], AND([Ref Loans].[Ref Borrowers] = [_THISROW].[Row ID], [Historical On Time] = TRUE)))
Lifetime On Time Rate	VC / Percent	OLAP only	-	Derived ratio or performance measure calculated from governed source values.	IF([Historical Charge Count] = 0, 0, [On Time Charge Count] / [Historical Charge Count])
Recent 30D Charge Count	VC / Number	OLAP only	-	Derived count used for workload, portfolio, or risk analysis.	COUNT(SELECT(Charges[Row ID], AND([Ref Loans].[Ref Borrowers] = [_THISROW].[Row ID], [Charge Date] >= TODAY() - 30, [Charge Date] < TODAY())))
Recent 30D On Time Count	VC / Number	OLAP only	-	Derived count used for workload, portfolio, or risk analysis.	COUNT(SELECT(Charges[Row ID], AND([Ref Loans].[Ref Borrowers] = [_THISROW].[Row ID], [Charge Date] >= TODAY() - 30, [Charge Date] < TODAY(), [Historical On Time] = TRUE)))
Recent 30D On Time Rate	VC / Percent	OLAP only	-	Derived ratio or performance measure calculated from governed source values.	IF([Recent 30D Charge Count] = 0, 0, [Recent 30D On Time Count] / [Recent 30D Charge Count])
Lifetime Late Payment Count	VC / Number	OLAP only	-	Derived count used for workload, portfolio, or risk analysis.	COUNT(SELECT(Charges[Row ID], AND([Ref Loans].[Ref Borrowers] = [_THISROW].[Row ID], [Historical Late] = TRUE)))
Recent 30D Late Payment Count	VC / Number	OLAP only	-	Derived count used for workload, portfolio, or risk analysis.	COUNT(SELECT(Charges[Row ID], AND([Ref Loans].[Ref Borrowers] = [_THISROW].[Row ID], [Charge Date] >= TODAY() - 30, [Charge Date] < TODAY(), [Historical Late] = TRUE)))
Maximum Days Late	VC / Number	OLAP only	-	Derived business-state or risk indicator.	MAX(LIST(0) + SELECT(Charges[Days Late], [Ref Loans].[Ref Borrowers] = [_THISROW].[Row ID]))
Last Late Payment Date	VC / Date	OLAP only	-	Derived business-state or risk indicator.	MAX(SELECT(Charges[Charge Date], AND([Ref Loans].[Ref Borrowers] = [_THISROW].[Row ID], [Days Late] > 0)))
Largest Successful Loan	VC / Price	OLAP only	-	Derived AppSheet value used by the app workflow, presentation, or reporting model.	MAX(LIST(0) + SELECT(Loans[Principal Amount], AND([Ref Borrowers] = [_THISROW].[Row ID], [Loan Status] = "ปิดยอดแล้ว", [Defaulted] = FALSE)))
Average Successful Loan Size	VC / Price	OLAP only	-	Derived date, time-window, age, or delinquency	IF(COUNT(SELECT(Loans[Row ID], AND([Ref Borrowers] = [_THISROW].[Row

Column	Kind / type	Availability	Flags	Purpose	Definition / expression
				measure.	ID], [Loan Status] = "ปิดยอดแล้ว", [Defaulted] = FALSE))) = 0, 0, AVERAGE(SELECT(Loans[Principal Amount], AND([Ref Borrowers] = [_THISROW].[Row ID], [Loan Status] = "ปิดยอดแล้ว", [Defaulted] = FALSE)))
Default Count	VC / Number	OLAP only	-	Derived count used for workload, portfolio, or risk analysis.	COUNT(SELECT(Loans[Row ID], AND([Ref Borrowers] = [_THISROW].[Row ID], [Defaulted] = TRUE)))
Lifetime Default Loss	VC / Price	OLAP only	-	Derived AppSheet value used by the app workflow, presentation, or reporting model.	SUM(SELECT(Loans[Default Loss Amount], AND([Ref Borrowers] = [_THISROW].[Row ID], [Defaulted] = TRUE)))
Portfolio Outstanding Principal	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	SUM(Loans[Outstanding Principal])
Portfolio Share Join Key	VC / Text	OLAP only	-	Derived grouping, ordering, or join value used by reporting or app presentation.	"Portfolio"
Daily Collection Status - Order	VC / Text	OLTP only	-	Derived grouping, ordering, or join value used by reporting or app presentation.	SWITCH([Daily Collection Status], "Not Paid", "1 - ยังไม่ชำระ Not Paid", "Partially Paid", "2 - ชำระบางส่วน Partially Paid", "Overdue", "3 - เกินกำหนด Overdue", "Fully Paid", "4 - ชำระครบ Fully Paid", "")
Borrower Name - Collection	VC / Text	OLTP only	-	Derived AppSheet value used by the app workflow, presentation, or reporting model.	CONCATENATE([Borrower Name], " - ", [Description])

6.2 Calendar

Source: Calendar Help Table / Sheet1 (Google Sheets)

Loaded in: OLAP only

Purpose: Google Sheets-backed date spine used by OLAP loan-assessment projections; not part of Money database.

Column	Kind / type	Availability	Flags	Purpose	Definition / expression
_RowNumber	Physical / Number	OLAP only	-	System-generated source-row ordinal; not a stable business key.	Stored source column
Date	Physical / Date	OLAP only	Key, Label	Stores the source date for Date.	Stored source column
Assessment Proposed Amount	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	LOOKUP("MAIN", "Loan Assessment", "Assessment ID", "Proposed Loan Amount")
Assessment Borrower	VC / Ref	OLAP only	-	Reference key that links this row to the configured parent table.	LOOKUP("MAIN", "Loan Assessment", "Assessment ID", "Ref Borrower")
Assessment Minimum Rate	VC / Percent	OLAP only	-	Derived ratio or performance measure calculated from governed source values.	LOOKUP("MAIN", "Loan Assessment", "Assessment ID", "Minimum Daily Profit Rate")
Projected Outstanding Principal	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	SUM(SELECT(Loans[Outstanding Principal], AND([Ref Borrowers] = [_THISROW].[Assessment Borrower], [Loan Status] = "ยังไม่ปิดยอด", OR([Loan Type] = "ดอกเบี้ยรายวัน", AND([Loan Type] = "กำหนดวันชำระ", [Due Date] > [_THISROW].[Date])))))
Projected Interest	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	SUM(SELECT(Loans[Total Interest Received], AND([Ref Borrowers] = [_THISROW].[Assessment Borrower], [Loan Status] = "ยังไม่ปิดยอด")))+(SUM(SELECT(Loans[Current Daily Interest], AND([Ref Borrowers] = [_THISROW].[Assessment Borrower], [Loan Status] = "ยังไม่ปิดยอด", [Loan Type] = "ดอกเบี้ยรายวัน"))) * IF([Date] > TODAY(), TOTALHOURS([Date] - TODAY()) / 24, 0))+SUM(SELECT(Loans[Fixed Interest], AND([Ref Borrowers] = [_THISROW].[Assessment Borrower], [Loan Status] = "ยังไม่ปิดยอด", [Loan Type] = "กำหนดวันชำระ", [Due Date] > TODAY(), [Due Date] <= [_THISROW].[Date])))
Candidate Day Index	VC / Decimal	OLAP only	-	Derived date, time-window, age, or delinquency measure.	TOTALHOURS([Date] - DATE("01/01/2000")) / 24
Assessment Principal Base	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	SUM(SELECT(Loans[Principal Amount], AND([Ref Borrowers] = [_THISROW].[Assessment Borrower], [Loan Status] = "ยังไม่ปิดยอด", OR([Loan Type] = "ดอกเบี้ยรายวัน", AND([Loan Type] = "กำหนดวันชำระ", [Due Date] > [_THISROW].[Date])))))
Assessment Principal Start Weight	VC / Decimal	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	SUM(SELECT(Loans[Principal Start Weight], AND([Ref Borrowers] = [_THISROW].[Assessment Borrower], [Loan Status] = "ยังไม่ปิดยอด", OR([Loan Type] = "ดอกเบี้ยรายวัน", AND([Loan Type] = "กำหนดวันชำระ", [Due Date] > [_THISROW].[Date])))))
Minimum Required Profit	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	[Assessment Minimum Rate]* ((([Candidate Day Index] + 1) * [Assessment Principal Base])- [Assessment Principal Start Weight])
Required Coverage	VC / Price	OLAP only	-	Derived ratio or performance measure calculated from governed source values.	[Projected Outstanding Principal]+[Assessment Proposed Amount]+[Minimum Required Profit]
Coverage Margin	VC / Price	OLAP only	-	Derived ratio or performance measure calculated from governed source values.	[Projected Interest] - [Required Coverage]

Column	Kind / type	Availability	Flags	Purpose	Definition / expression
Eligible	VC / Yes/No	OLAP only	-	Derived business-state or risk indicator.	AND([Date] >= TODAY(), [Projected Interest] >= [Required Coverage])
Year	VC / Number	OLAP only	-	Derived date, time-window, age, or delinquency measure.	YEAR([Date])
Month	VC / Number	OLAP only	-	Derived date, time-window, age, or delinquency measure.	MONTH([Date])
Month Name	VC / Text	OLAP only	-	Derived date, time-window, age, or delinquency measure.	TEXT([Date], "MMMM")
Year Month	VC / Text	OLAP only	-	Derived date, time-window, age, or delinquency measure.	TEXT([Date], "YYYY-MM")
Week	VC / Number	OLAP only	-	Derived date, time-window, age, or delinquency measure.	WEEKNUM([Date])
Year Week	VC / Text	OLAP only	-	Derived date, time-window, age, or delinquency measure.	CONCATENATE(YEAR([Date]), "-W", RIGHT("0" & WEEKNUM([Date]), 2))
Week Start	VC / Date	OLAP only	-	Derived date, time-window, age, or delinquency measure.	[Date] - MOD(WEEKDAY([Date]) + 5, 7)
Quarter	VC / Text	OLAP only	-	Derived AppSheet value used by the app workflow, presentation, or reporting model.	CONCATENATE("Q", CEILING(MONTH([Date]) / 3))
Day of Week	VC / Text	OLAP only	-	Derived date, time-window, age, or delinquency measure.	TEXT([Date], "dddd")
Day of Month	VC / Number	OLAP only	-	Derived date, time-window, age, or delinquency measure.	DAY([Date])

6.3 Cash Pool Contributions

Source: Money database / Cash Pool Contributions (AppSheet Database)

Loaded in: OLAP only

Purpose: Partner capital-contribution movements that build the lendable cash pool.

Column	Kind / type	Availability	Flags	Purpose	Definition / expression
_RowNumber	Physical / Number	OLAP only	-	System-generated source-row ordinal; not a stable business key.	Stored source column
Row ID	Physical / Text	OLAP only	Key	Immutable row key generated by the source database.	Stored source column; initial value: UNIQUEID("PackedUUID")
Ref Partner	Physical / Ref	OLAP only	-	Reference key that links this row to the configured parent table.	Stored source column
Contribution Date	Physical / Date	OLAP only	Label	Stores the source date for Contribution Date.	Stored source column; initial value: TODAY()
Transaction Type	Physical / Enum	OLAP only	-	Stores the authoritative source value for Transaction Type.	Stored source column; initial value: "Contribution"
Amount	Physical / Price	OLAP only	-	Stores the financially material source value for Amount.	Stored source column
Notes	Physical / Text	OLAP only	-	Stores the authoritative source value for Notes.	Stored source column
Net Contribution	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	IF([Transaction Type] = "Contribution", [Amount], -[Amount])
Transaction Type TE	VC / Text	OLAP only	-	Presentation-oriented label or bilingual display value.	IF([Transaction Type] = "Contribution", "เพิ่มเงิน - Contribution", "ถอนเงิน - Withdrawal")

6.4 Charges

Source: Money database / Charges (AppSheet Database)

Loaded in: OLTP + OLAP

Purpose: Scheduled amounts due and collection state at charge level, including principal and interest allocation.

Column	Kind / type	Availability	Flags	Purpose	Definition / expression
_RowNumber	Physical / Number	OLTP + OLAP	-	System-generated source-row ordinal; not a stable business key.	Stored source column
Row ID	Physical / Text	OLTP + OLAP	Key	Immutable row key generated by the source database.	Stored source column; initial value: UNIQUEID("PackedUUID")
Ref Loans	Physical / Ref	OLTP + OLAP	-	Reference key that links this row to the configured parent table.	Stored source column
Charge Date	Physical / Date	OLTP + OLAP	Label	Stores the source date for Charge Date.	Stored source column; initial value: TODAY()
Interest Due	Physical / Price	OLTP + OLAP	-	Stores the financially material source value for Interest Due.	Stored source column; initial value: 0
Principal Due	Physical / Price	OLTP + OLAP	-	Stores the financially material source value for Principal Due.	Stored source column; initial value: 0
Notes	Physical / LongText	OLTP + OLAP	-	Stores the authoritative source value for Notes.	Stored source column
Amount Due	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	[Principal Due] + [Interest Due]
Borrower Name	VC / Name	OLTP + OLAP	-	Derived AppSheet value used by the app workflow, presentation, or reporting model.	[Ref Loans].[Borrower Name]
Related Repayments	VC / List	OLTP + OLAP	-	Reverse-reference list of child rows that point to this record.	REF_ROWS("Repayments", "Ref Charges")
Payment Status	VC / Text	OLTP + OLAP	-	Derived business-state or risk indicator.	IFS([Ref Loans].[Loan Status] = "ปิดยอดแล้ว", "ชำระแล้ว", [Total Paid] = 0, "รอชำระ", [Amount Remaining] > 0, "ชำระบางส่วน", TRUE, "ชำระแล้ว")
Payment Date	VC / Date	OLTP + OLAP	-	Derived date, time-window, age, or delinquency measure.	IF([Payment Status] = "ชำระแล้ว", MAX([Related Repayments][Payment Date]), "")
Borrower Name TE	VC / Text	OLTP + OLAP	-	Presentation-oriented label or bilingual display value.	[Ref Loans].[Borrower Name TE]
Payment Status TE	VC / Text	OLTP + OLAP	-	Presentation-oriented label or bilingual display value.	IFS([Payment Status] = "ชำระแล้ว", "ชำระแล้ว / Payment Received", [Payment Status] = "รอชำระ", "รอชำระ / Pending Payment", TRUE, "ชำระบางส่วน / Partial Payment")
Principal Paid	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	SUM([Related Repayments][Principal Paid])
Interest Paid	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	SUM([Related Repayments][Interest Paid])
Total Paid	VC / Price	OLTP + OLAP	-	Derived AppSheet value used by the app workflow, presentation, or reporting model.	[Principal Paid] + [Interest Paid]
Principal Remaining	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	[Principal Due] - [Principal Paid]

Column	Kind / type	Availability	Flags	Purpose	Definition / expression
Interest Remaining	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	[Interest Due] - [Interest Paid]
Amount Remaining	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	[Principal Remaining] + [Interest Remaining]
Last Payment Date	VC / Date	OLAP only	-	Derived date, time-window, age, or delinquency measure.	IF(COUNT([Related Repayments]) > 0, MAX([Related Repayments][Payment Date]), "")
First Payment Date	VC / Date	OLAP only	-	Derived date, time-window, age, or delinquency measure.	MIN([Related Repayments][Payment Date])
Days Late	VC / Number	OLAP only	-	Derived business-state or risk indicator.	IFS(AND([Payment Status] = "ชำระแล้ว", ISNOTBLANK([Payment Date])), MAX(LIST(0, FLOOR(TOTALHOURS([Payment Date] - [Charge Date]) / 24))), [Charge Date] < TODAY(), FLOOR(TOTALHOURS(TODAY() - [Charge Date]) / 24), TRUE, 0)
Historical On Time	VC / Yes/No	OLAP only	-	Derived business-state or risk indicator.	AND([Charge Date] < TODAY(), [Payment Status] = "ชำระแล้ว", [Days Late] = 0)
Historical Late	VC / Yes/No	OLAP only	-	Derived business-state or risk indicator.	AND([Charge Date] < TODAY(), [Days Late] > 0)
Payment Count	VC / Number	OLTP only	-	Derived count used for workload, portfolio, or risk analysis.	COUNT([Related Repayments])
Days Outstanding	VC / Number	OLTP only	-	Derived date, time-window, age, or delinquency measure.	FLOOR(TOTALHOURS(IF([Payment Status] = "ชำระแล้ว", [Payment Date] - [Charge Date], TODAY() - [Charge Date])) / 24)

6.5 Daily Analytics

Source: Money database / Daily Analytics (AppSheet Database)

Loaded in: OLAP only

Purpose: Persisted daily portfolio snapshot fact table for historical reporting and governed feeds.

Column	Kind / type	Availability	Flags	Purpose	Definition / expression
_RowNumber	Physical / Number	OLAP only	-	System-generated source-row ordinal; not a stable business key.	Stored source column
Row ID	Physical / Text	OLAP only	Key	Immutable row key generated by the source database.	Stored source column; initial value: UNIQUEID("PackedUUID")
Snapshot Date	Physical / Date	OLAP only	Label	Stores the source date for Snapshot Date.	Stored source column; initial value: TODAY()
Generated At	Physical / DateTime	OLAP only	-	Stores the authoritative source value for Generated At.	Stored source column; initial value: NOW()
Model Version	Physical / Number	OLAP only	-	Stores the authoritative source value for Model Version.	Stored source column
Principal Issued	Physical / Price	OLAP only	-	Stores the financially material source value for Principal Issued.	Stored source column
Loans Issued	Physical / Number	OLAP only	-	Stores the authoritative source value for Loans Issued.	Stored source column
Principal Returned	Physical / Price	OLAP only	-	Stores the financially material source value for Principal Returned.	Stored source column
Interest Received	Physical / Price	OLAP only	-	Stores the financially material source value for Interest Received.	Stored source column
Repayments Count	Physical / Number	OLAP only	-	Stores the authoritative source value for Repayments Count.	Stored source column
Outstanding Principal EOD	Physical / Price	OLAP only	-	Stores the financially material source value for Outstanding Principal EOD.	Stored source column
Active Loans EOD	Physical / Number	OLAP only	-	Stores the authoritative source value for Active Loans EOD.	Stored source column
Active Borrowers EOD	Physical / Number	OLAP only	-	Stores the authoritative source value for Active Borrowers EOD.	Stored source column
Pending Charges EOD	Physical / Price	OLAP only	-	Stores the authoritative source value for Pending Charges EOD.	Stored source column
Total Cash Pool EOD	Physical / Price	OLAP only	-	Stores the financially material source value for Total Cash Pool EOD.	Stored source column
Available Cash EOD	Physical / Price	OLAP only	-	Stores the financially material source value for Available Cash EOD.	Stored source column
Unsettled Profit EOD	Physical / Price	OLAP only	-	Stores the financially material source value for Unsettled Profit EOD.	Stored source column
Week	VC / Text	OLAP only	-	Derived date, time-window, age, or delinquency measure.	CONCATENATE(YEAR([Snapshot Date]), "-W", WEEKNUM([Snapshot Date]))
Month	VC / Text	OLAP only	-	Derived date, time-window, age, or delinquency	CONCATENATE(YEAR([Snapshot Date]), "-", RIGHT("0" &

Column	Kind / type	Availability	Flags	Purpose	Definition / expression
				measure.	MONTH([Snapshot Date]), 2))
Daily Capital Employed	VC / Price	OLAP only	-	Derived AppSheet value used by the app workflow, presentation, or reporting model.	[Outstanding Principal EOD]+[Principal Returned]
Daily Capital Yield	VC / Percent	OLAP only	-	Derived ratio or performance measure calculated from governed source values.	IF([Daily Capital Employed] > 0, [Interest Received] / [Daily Capital Employed], 0)
Portfolio Share Join Key	VC / Text	OLAP only	-	Derived grouping, ordering, or join value used by reporting or app presentation.	"Portfolio"
Portfolio Slice	VC / Text	OLAP only	-	Derived grouping, ordering, or join value used by reporting or app presentation.	IF(MOD(DAY([Snapshot Date]), 2) = 0, "Selected borrower", "Remaining")

6.6 Loan Assessment

Source: Money database / Loan Assessment (AppSheet Database)

Loaded in: OLAP only

Purpose: Input/control record for evaluating a proposed loan against projected coverage and return.

Column	Kind / type	Availability	Flags	Purpose	Definition / expression
_RowNumber	Physical / Number	OLAP only	-	System-generated source-row ordinal; not a stable business key.	Stored source column
Row ID	Physical / Text	OLAP only	Key	Immutable row key generated by the source database.	Stored source column; initial value: UNIQUEID("PackedUUID")
Assessment ID	Physical / Text	OLAP only	Label	Stores the authoritative source value for Assessment ID.	Stored source column
Ref Borrower	Physical / Ref	OLAP only	-	Reference key that links this row to the configured parent table.	Stored source column
Proposed Loan Amount	Physical / Price	OLAP only	-	Stores the financially material source value for Proposed Loan Amount.	Stored source column
Minimum Daily Profit Rate	Physical / Percent	OLAP only	-	Stores the financially material source value for Minimum Daily Profit Rate.	Stored source column
Current Outstanding Principal	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	SUM(SELECT(Loans[Outstanding Principal], AND([Ref Borrowers] = [_THISROW].[Ref Borrower], [Loan Status] = "ยังไม่ปิดยอด")))
Interest Earned from Active Loans	VC / Price	OLAP only	-	Derived count used for workload, portfolio, or risk analysis.	SUM(SELECT(Loans[Total Interest Received], AND([Ref Borrowers] = [_THISROW].[Ref Borrower], [Loan Status] = "ยังไม่ปิดยอด")))
Earliest Eligible Date	VC / Date	OLAP only	-	Derived business-state or risk indicator.	MIN(SELECT(Calendar[Date], AND([Date] >= TODAY(), [Eligible] = TRUE)))
Days Until Eligible	VC / Number	OLAP only	-	Derived business-state or risk indicator.	IF(ISBLANK([Earliest Eligible Date]), "", TOTALHOURS([Earliest Eligible Date] - TODAY()) / 24)
Projected Interest at Eligible Date	VC / Price	OLAP only	-	Derived business-state or risk indicator.	LOOKUP([Earliest Eligible Date], "Calendar", "Date", "Projected Interest")
Outstanding Principal at Eligible Date	VC / Price	OLAP only	-	Derived business-state or risk indicator.	LOOKUP([Earliest Eligible Date], "Calendar", "Date", "Projected Outstanding Principal")
Minimum Profit at Eligible Date	VC / Price	OLAP only	-	Derived business-state or risk indicator.	LOOKUP([Earliest Eligible Date], "Calendar", "Date", "Minimum Required Profit")
Required Coverage at Eligible Date	VC / Price	OLAP only	-	Derived ratio or performance measure calculated from governed source values.	LOOKUP([Earliest Eligible Date], "Calendar", "Date", "Required Coverage")
Safety Margin	VC / Price	OLAP only	-	Derived AppSheet value used by the app workflow, presentation, or reporting model.	LOOKUP([Earliest Eligible Date], "Calendar", "Date", "Coverage Margin")
Current Coverage Margin	VC / Price	OLAP only	-	Derived ratio or performance measure calculated from governed source values.	LOOKUP(TODAY(), "Calendar", "Date", "Coverage Margin")
Coverage Ratio at Eligible Date	VC / Percent	OLAP only	-	Derived ratio or performance measure calculated from governed source values.	IF([Required Coverage at Eligible Date] > 0, [Projected Interest at Eligible Date] / [Required Coverage at Eligible Date], 0)
Recommendation Status	VC / Text	OLAP only	-	Derived business-state or risk indicator.	IFS(ISBLANK([Earliest Eligible Date]), "ยังไม่ผ่านเกณฑ์ - Not Eligible", [Earliest Eligible Date] <= TODAY(), "ปล่อยกู้ได้ตอนนี้ - Eligible Now", TRUE, "ควรรอก่อน - Wait")

Column	Kind / type	Availability	Flags	Purpose	Definition / expression
Recommendation	VC / Text	OLAP only	-	Derived AppSheet value used by the app workflow, presentation, or reporting model.	IFS(ISBLANK([Ref Borrower]), "กรุณาเลือกผู้กู้ / Select a borrower", OR(ISBLANK([Proposed Loan Amount]), [Proposed Loan Amount] <= 0), "กรุณากรอกวงเงินกู้ที่เสนอ / Enter the proposed loan amount", OR(ISBLANK([Minimum Daily Profit Rate]), [Minimum Daily Profit Rate] <= 0), "กรุณากรอกอัตรากำไรขั้นต่ำ / Enter the minimum profit rate", ISBLANK([Earliest Eligible Date]), "ไม่พบวันที่ผ่านเกณฑ์ภายในช่วงเวลาที่ประเมิน / No eligible date found within the forecast period", [Earliest Eligible Date] = TODAY(), "สามารถปล่อยกู้ได้วันนี้ / Eligible to issue the loan today", TRUE, CONCATENATE("วันที่แนะนำให้ปล่อยกู้ได้เร็วที่สุด / Earliest recommended issue date: ", TEXT([Earliest Eligible Date], "DD/MM/YYYY")))
Header - Current Position	VC / Text	OLAP only	-	Presentation-oriented label or bilingual display value.	"—— สถานะปัจจุบัน / Current Position ——"
Header - Recommendation	VC / Text	OLAP only	-	Presentation-oriented label or bilingual display value.	"—— คำแนะนำ / Recommendation ——"
Header - Calculation	VC / Text	OLAP only	-	Presentation-oriented label or bilingual display value.	"—— รายละเอียดการคำนวณ / Calculation ——"

6.7 Loans

Source: Money database / Loans (AppSheet Database)

Loaded in: OLTP + OLAP

Purpose: Loan contracts, commercial terms, lifecycle state, balances, and loan-level risk metrics.

Column	Kind / type	Availability	Flags	Purpose	Definition / expression
_RowNumber	Physical / Number	OLTP + OLAP	-	System-generated source-row ordinal; not a stable business key.	Stored source column
Row ID	Physical / Text	OLTP + OLAP	Key	Immutable row key generated by the source database.	Stored source column; initial value: UNIQUEID("PackedUUID")
Loan Date	Physical / Date	OLTP + OLAP	-	Stores the source date for Loan Date.	Stored source column; initial value: TODAY()
Close Date	Physical / Date	OLTP + OLAP	-	Stores the source date for Close Date.	Stored source column; initial value: TODAY()
Principal Amount	Physical / Price	OLTP + OLAP	-	Stores the financially material source value for Principal Amount.	Stored source column
Due Date	Physical / Date	OLTP + OLAP	-	Stores the source date for Due Date.	Stored source column; initial value: TODAY()
Ref Borrowers	Physical / Ref	OLTP + OLAP	-	Reference key that links this row to the configured parent table.	Stored source column
Loan Arrangement	Physical / LongText	OLTP + OLAP	-	Stores the authoritative source value for Loan Arrangement.	Stored source column
Loan Status	Physical / Enum	OLTP + OLAP	-	Stores the controlled workflow state for Loan Status.	Stored source column; initial value: "ยังไม่ปิดยอด"
Borrower Name	Physical / Name	OLTP + OLAP	-	Stores the authoritative source value for Borrower Name.	Stored source column
Loan Type	Physical / Enum	OLTP + OLAP	-	Stores the authoritative source value for Loan Type.	Stored source column
Current Daily Interest	Physical / Price	OLTP + OLAP	-	Stores the financially material source value for Current Daily Interest.	Stored source column
Fixed Interest	Physical / Price	OLTP + OLAP	-	Stores the financially material source value for Fixed Interest.	Stored source column
Transfer Fee	Physical / Price	OLTP + OLAP	-	Stores the financially material source value for Transfer Fee.	Stored source column; initial value: 0
Interest Payment Interval	Physical / Number	OLTP + OLAP	-	Stores the financially material source value for Interest Payment Interval.	Stored source column; initial value: 1
Created By	Physical / Email	OLTP + OLAP	-	Stores the authoritative source value for Created By.	Stored source column; initial value: USEREMAIL()
Closed By	Physical / Email	OLTP + OLAP	-	Stores the authoritative source value for Closed By.	Stored source column
Defaulted	Physical / Yes/No	OLTP + OLAP	-	Stores the authoritative source value for Defaulted.	Stored source column; initial value: FALSE
Default Loss Amount	Physical / Price	OLTP + OLAP	-	Stores the financially material source value for Default Loss Amount.	Stored source column; initial value: 0
Related Repayments	VC / List	OLTP + OLAP	-	Reverse-reference list of child rows that point to this record.	REF_ROWS("Repayments", "Ref Loans")

Column	Kind / type	Availability	Flags	Purpose	Definition / expression
Total Interest Received	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	SUM([Related Repayments][Interest Paid])
Total Principal Received	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	SUM([Related Repayments][Principal Paid])
Outstanding Principal	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	[Principal Amount] - [Total Principal Received]
Total Amount Received	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	SUM([Related Repayments][Amount Received])
Statistics Header	VC / Text	OLTP + OLAP	-	Presentation-oriented label or bilingual display value.	"📊 สถิติการกู้ / Statistics"
Borrower Total Interest	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	SUM(SELECT(Loans[Total Interest Received], [Ref Borrowers] = [_THISROW].[Ref Borrowers]))
Related Charges	VC / List	OLTP + OLAP	-	Reverse-reference list of child rows that point to this record.	REF_ROWS("Charges", "Ref Loans")
Loan Key	VC / Text	OLTP + OLAP	Label	Derived AppSheet value used by the app workflow, presentation, or reporting model.	CONCATENATE([Borrower Name], "-", TEXT([Principal Amount]), "-", TEXT([Loan Date], "DD/MM"), "-", TEXT([Effective Daily Interest Rate]))
Expected Interest This Month	VC / Price	OLTP + OLAP	-	Derived date, time-window, age, or delinquency measure.	IF([Loan Status] = "ปิดยอดแล้ว", 0, IFS([Loan Type] = "ดอกเบี้ยรายวัน", [Current Daily Interest] * (DAY(EOMONTH(TODAY(), 0)) - DAY(TODAY())), [Loan Type] = "กำหนดวันชำระ", IF(AND([Due Date] > TODAY(), MONTH([Due Date]) = MONTH(TODAY()), YEAR([Due Date]) = YEAR(TODAY())), [Fixed Interest], 0)))
Effective Daily Interest Rate	VC / Percent	OLTP + OLAP	-	Derived ratio or performance measure calculated from governed source values.	IFS([Loan Type] = "ดอกเบี้ยรายวัน", [Current Daily Interest], [Loan Type] = "กำหนดวันชำระ", [Fixed Interest] / (HOUR([Due Date] - [Loan Date]) / 24 + 1))/ [Principal Amount]
Principal Recovered by Interest %	VC / Percent	OLTP + OLAP	-	Derived ratio or performance measure calculated from governed source values.	IF([Principal Amount] > 0, [Total Interest Received] / [Principal Amount], 0)
Principal Days MTD	VC / Number	OLTP + OLAP	-	Derived date, time-window, age, or delinquency measure.	IF([Loan Date] > TODAY(), 0, [Principal Amount] * IF(AND(MONTH([Loan Date]) = MONTH(TODAY()), YEAR([Loan Date]) = YEAR(TODAY())), DAY(TODAY()) - DAY([Loan Date]) + 1, DAY(TODAY())))
Borrower Name TE	VC / Text	OLTP + OLAP	-	Presentation-oriented label or bilingual display value.	[Ref Borrowers].[Borrower Name TE]
Loan Status TE	VC / Text	OLTP + OLAP	-	Presentation-oriented label or bilingual display value.	IF([Loan Status] = "ปิดยอดแล้ว", "ปิดยอดแล้ว / Closed", "ยังไม่ปิดยอด / Active")
Loan Type TE	VC / Text	OLTP + OLAP	-	Presentation-oriented label or bilingual display value.	IF([Loan Type] = "กำหนดวันชำระ", "กำหนดวันชำระ / Fixed Due Date", "ดอกเบี้ยรายวัน / Daily Interest")
Loan Start Day Index	VC / Decimal	OLAP only	-	Derived AppSheet value used by the app workflow, presentation, or reporting model.	TOTALHOURS([Loan Date] - DATE("01/01/2000")) / 24
Principal Start Weight	VC / Decimal	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	[Principal Amount] * [Loan Start Day Index]
Loan End Date	VC / Date	OLAP only	-	Derived date, time-window, age, or delinquency measure.	[Close Date]
Loan Duration	VC / Number	OLTP + OLAP	-	Derived date, time-window, age, or delinquency measure.	OLTP: IF(ISNOTBLANK([Close Date]), FLOOR(TOTALHOURS([Close Date] - [Loan Date]) / 24) + 1, "")

Column	Kind / type	Availability	Flags	Purpose	Definition / expression
					OLAP: IF(ISNOTBLANK([Loan End Date]), TOTALHOURS([Loan End Date] - [Loan Date]) / 24 + 1, "")
30D Principal Days	VC / Decimal	OLAP only	-	Derived date, time-window, age, or delinquency measure.	IF([Loan Date] > TODAY(), 0, [Principal Amount] * (FLOOR(TOTALHOURS(TODAY() - MAX(LIST([Loan Date], TODAY() - 29))) / 24) + 1))
Loan Age Days	VC / Number	OLAP only	-	Derived date, time-window, age, or delinquency measure.	FLOOR(TOTALHOURS(TODAY() - [Loan Date]) / 24)
Loan Age Bucket	VC / Text	OLAP only	-	Derived grouping, ordering, or join value used by reporting or app presentation.	IFS([Loan Age Days] <= 10, "0-10 Days", [Loan Age Days] <= 20, "11-20 Days", [Loan Age Days] <= 30, "21-30 Days", [Loan Age Days] <= 60, "31-60 Days", TRUE, "60+ Days")
Defaulted Flag	VC / Yes/No	OLAP only	-	Derived business-state or risk indicator.	IF([Defaulted], TRUE, FALSE)
Maximum Days Late	VC / Number	OLAP only	-	Derived business-state or risk indicator.	MAX(LIST(0) + SELECT(Charges[Days Late], [Ref Loans] = [_THISROW].[Row ID]))
Last Interest Charge Date	VC / Date	OLTP only	-	Derived date, time-window, age, or delinquency measure.	MAX([Related Charges][Charge Date])
Accrued Interest Days	VC / Number	OLTP only	-	Derived date, time-window, age, or delinquency measure.	IF(ISBLANK([Last Interest Charge Date]), TOTALHOURS(TODAY() - [Loan Date])/24, TOTALHOURS(TODAY() - [Last Interest Charge Date])/24)
Accrued Interest	VC / Price	OLTP only	-	Derived financial aggregate or balance for operational or analytical use.	IF([Loan Type] = "ดอกเบี้ยรายวัน", [Current Daily Interest] * [Accrued Interest Days], 0)

6.8 Partners

Source: Money database / Partners (AppSheet Database)

Loaded in: OLTP + OLAP

Purpose: Partner master records used for capital participation and settlement reporting.

Column	Kind / type	Availability	Flags	Purpose	Definition / expression
_RowNumber	Physical / Number	OLTP + OLAP	-	System-generated source-row ordinal; not a stable business key.	Stored source column
Row ID	Physical / Text	OLTP + OLAP	Key	Immutable row key generated by the source database.	Stored source column; initial value: UNIQUEID("PackedUUID")
Partner Name	Physical / Name	OLTP + OLAP	Label	Stores the authoritative source value for Partner Name.	Stored source column
Partner Role	Physical / Enum	OLTP + OLAP	-	Stores the authoritative source value for Partner Role.	Stored source column
Email	Physical / Email	OLTP + OLAP	-	Stores the authoritative source value for Email.	Stored source column
Related Cash Pool Contributions	VC / List	OLAP only	-	Reverse-reference list of child rows that point to this record.	REF_ROWS("Cash Pool Contributions", "Ref Partner")
Related Settlements	VC / List	OLAP only	-	Reverse-reference list of child rows that point to this record.	REF_ROWS("Settlements", "Ref Partner")
Current Pool Contribution	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	SUM(SELECT(Cash Pool Contributions[Net Contribution], [Ref Partner] = [_THISROW].[Row ID]))
Current Pool Share	VC / Percent	OLAP only	-	Derived ratio or performance measure calculated from governed source values.	IF(SUM(Cash Pool Contributions[Net Contribution]) > 0, [Current Pool Contribution] / SUM(Cash Pool Contributions[Net Contribution]), 0)
Total Profit Earned	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	IFS([Partner Role] = "A", SUM(Repayments[Partner A Profit]), [Partner Role] = "B", SUM(Repayments[Partner B Profit]))
Total Settlement Created	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	SUM([Related Settlements][Amount])
Completed Settlement	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	SUM(SELECT(Settlements[Amount], AND([Ref Partner] = [_THISROW].[Row ID], [Status] = "Completed")))
Pending Settlement	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	SUM(SELECT(Settlements[Amount], AND([Ref Partner] = [_THISROW].[Row ID], [Status] = "Pending")))
Available to Settle	VC / Price	OLAP only	-	Derived AppSheet value used by the app workflow, presentation, or reporting model.	[Total Profit Earned] - [Total Settlement Created]
Summary - Available Pending Settlement	VC / Text	OLAP only	-	Presentation-oriented label or bilingual display value.	IF([Pending Settlement] > 0, CONCATENATE("รอโอน / Pending Settlement: ", TEXT([Pending Settlement])), CONCATENATE("พร้อมโอน / Available to Settle: ", TEXT([Available to Settle])))
Summary - Completed Settlement	VC / Text	OLAP only	-	Presentation-oriented label or bilingual display value.	CONCATENATE("โอนแล้ว / Completed Settlement: ", TEXT([Completed Settlement]))
Summary - Total Profit Earned	VC / Text	OLAP only	-	Presentation-oriented label or bilingual display value.	CONCATENATE("กำไรรวมที่ได้รับ / Total Profit: ", TEXT([Total Profit Earned]))

6.9 Repayments

Source: Money database / Repayments (AppSheet Database)

Loaded in: OLTP + OLAP

Purpose: Posted payment transactions and their allocation to loans and charges.

Column	Kind / type	Availability	Flags	Purpose	Definition / expression
_RowNumber	Physical / Number	OLTP + OLAP	-	System-generated source-row ordinal; not a stable business key.	Stored source column
Row ID	Physical / Text	OLTP + OLAP	Key	Immutable row key generated by the source database.	Stored source column; initial value: UNIQUEID("PackedUUID")
Payment Date	Physical / Date	OLTP + OLAP	Label	Stores the source date for Payment Date.	Stored source column; initial value: TODAY()
Principal Paid	Physical / Price	OLTP + OLAP	-	Stores the financially material source value for Principal Paid.	Stored source column; initial value: 0
Interest Paid	Physical / Price	OLTP + OLAP	-	Stores the financially material source value for Interest Paid.	Stored source column; initial value: 0
Notes	Physical / LongText	OLTP + OLAP	-	Stores the authoritative source value for Notes.	Stored source column
Ref Loans	Physical / Ref	OLTP + OLAP	-	Reference key that links this row to the configured parent table.	Stored source column
Ref Charges	Physical / Ref	OLTP + OLAP	-	Reference key that links this row to the configured parent table.	Stored source column
Created By	Physical / Email	OLTP + OLAP	-	Stores the authoritative source value for Created By.	Stored source column; initial value: USEREMAIL()
Borrower Name	VC / Name	OLTP + OLAP	-	Derived AppSheet value used by the app workflow, presentation, or reporting model.	[Ref Loans].[Borrower Name]
Principal Returned Days MTD	VC / Number	OLTP + OLAP	-	Derived ratio or performance measure calculated from governed source values.	IF(OR([Principal Paid] <= 0, [Payment Date] > TODAY()), 0, [Principal Paid] * IF(AND(MONTH([Payment Date]) = MONTH(TODAY()), YEAR([Payment Date]) = YEAR(TODAY())), DAY(TODAY()) - DAY([Payment Date]) + 1, DAY(TODAY())))
Pool Total at Payment Date	VC / Price	OLAP only	-	Derived date, time-window, age, or delinquency measure.	SUM(SELECT(Cash Pool Contributions[Net Contribution], [Contribution Date] <= [_THISROW].[Payment Date]))
Partner A Contribution at Payment Date	VC / Price	OLAP only	-	Derived date, time-window, age, or delinquency measure.	SUM(SELECT(Cash Pool Contributions[Net Contribution], AND([Contribution Date] <= [_THISROW].[Payment Date], [Ref Partner].[Partner Role] = "A")))
Partner B Contribution at Payment Date	VC / Price	OLAP only	-	Derived date, time-window, age, or delinquency measure.	SUM(SELECT(Cash Pool Contributions[Net Contribution], AND([Contribution Date] <= [_THISROW].[Payment Date], [Ref Partner].[Partner Role] = "B")))
Partner A Profit Share	VC / Percent	OLAP only	-	Derived ratio or performance measure calculated from governed source values.	IF([Pool Total at Payment Date] > 0, [Partner A Contribution at Payment Date] / [Pool Total at Payment Date], 0)
Partner B Profit Share	VC / Percent	OLAP only	-	Derived ratio or performance measure calculated from governed source values.	IF([Pool Total at Payment Date] > 0, [Partner B Contribution at Payment Date] / [Pool Total at Payment Date], 0)
Partner A Profit	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	[Interest Paid] * [Partner A Profit Share]

Column	Kind / type	Availability	Flags	Purpose	Definition / expression
Partner B Profit	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	[Interest Paid] * [Partner B Profit Share]
Header - Profit	VC / Text	OLAP only	-	Presentation-oriented label or bilingual display value.	การแบ่งกำไร
Borrower Name TE	VC / Text	OLTP + OLAP	-	Presentation-oriented label or bilingual display value.	[Ref Loans].[Borrower Name TE]
30D Principal Reduction Days	VC / Decimal	OLAP only	-	Derived date, time-window, age, or delinquency measure.	IF(OR([Payment Date] < TODAY() - 29, [Payment Date] > TODAY()), 0, [Principal Paid] * (FLOOR(TOTALHOURS(TODAY() - [Payment Date]) / 24) + 1))
Amount Received	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	[Principal Paid] + [Interest Paid]

6.10 Settlements

Source: Money database / Settlements (AppSheet Database)

Loaded in: OLAP only

Purpose: Auditable partner settlement records and settlement status.

Column	Kind / type	Availability	Flags	Purpose	Definition / expression
_RowNumber	Physical / Number	OLAP only	-	System-generated source-row ordinal; not a stable business key.	Stored source column
Row ID	Physical / Text	OLAP only	Key	Immutable row key generated by the source database.	Stored source column; initial value: UNIQUEID("PackedUUID")
Settlement Date	Physical / Date	OLAP only	Label	Stores the source date for Settlement Date.	Stored source column; initial value: TODAY()
Ref Partner	Physical / Ref	OLAP only	-	Reference key that links this row to the configured parent table.	Stored source column
Amount	Physical / Price	OLAP only	-	Stores the financially material source value for Amount.	Stored source column
Status	Physical / Enum	OLAP only	-	Stores the controlled workflow state for Status.	Stored source column; initial value: "Pending"
Transfer Date	Physical / Date	OLAP only	-	Stores the source date for Transfer Date.	Stored source column
Notes	Physical / LongText	OLAP only	-	Stores the authoritative source value for Notes.	Stored source column

6.11 Statistics

Source: Money database / Statistics (AppSheet Database)

Loaded in: OLTP + OLAP

Purpose: Small KPI/control table used to expose app-level operational and reporting aggregates.

Column	Kind / type	Availability	Flags	Purpose	Definition / expression
_RowNumber	Physical / Number	OLTP + OLAP	-	System-generated source-row ordinal; not a stable business key.	Stored source column
Row ID	Physical / Text	OLTP + OLAP	Key	Immutable row key generated by the source database.	Stored source column; initial value: UNIQUEID("PackedUUID")
Statistics ID	Physical / Text	OLTP + OLAP	Label	Stores the authoritative source value for Statistics ID.	Stored source column
Total Profit	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	SUM(Repayments[Interest Paid])
Outstanding Principal	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	SUM(Loans[Outstanding Principal])
Active Loans	VC / Number	OLAP only	-	Derived count used for workload, portfolio, or risk analysis.	COUNT(SELECT(Loans[Row ID], [Loan Status] = "ยังไม่ปิดยอด"))
Today's Profit	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	SUM(SELECT(Repayments[Interest Paid], [Payment Date] = TODAY()))
Yesterday's Profit	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	SUM(SELECT(Repayments[Interest Paid], [Payment Date] = TODAY() - 1))
Realized Profit MTD	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	SUM(SELECT(Repayments[Interest Paid], AND(MONTH([Payment Date]) = MONTH(TODAY()), YEAR([Payment Date]) = YEAR(TODAY()))))
Projected Monthly Profit	VC / Price	OLTP + OLAP	-	Derived date, time-window, age, or delinquency measure.	[Realized Profit MTD]+[Today's Outstanding Interest]+[Forecast Remaining Interest]
Pending Payment Amount	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	SUM(Charges[Amount Remaining])
Forecast Remaining Interest	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	SUM(Loans[Expected Interest This Month])
Today's Outstanding Interest	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	SUM(Charges[Interest Remaining])
Net Principal Days MTD	VC / Number	OLTP + OLAP	-	Derived date, time-window, age, or delinquency measure.	SUM(Loans[Principal Days MTD]) - SUM(Repayments[Principal Returned Days MTD])
Forecast Remaining Principal Days	VC / Number	OLTP + OLAP	-	Derived date, time-window, age, or delinquency measure.	[Outstanding Principal] * (DAY(EOMONTH(TODAY(), 0)) - DAY(TODAY()))
Forecast Average Daily Outstanding Principal	VC / Price	OLTP + OLAP	-	Derived date, time-window, age, or delinquency measure.	([Net Principal Days MTD] + [Forecast Remaining Principal Days])/ DAY(EOMONTH(TODAY(), 0))
Forecast Monthly Return on ADO	VC / Percent	OLTP + OLAP	-	Derived ratio or performance measure calculated from governed source values.	IF([Forecast Average Daily Outstanding Principal] > 0, [Projected Monthly Profit] / [Forecast Average Daily Outstanding Principal], 0)

Column	Kind / type	Availability	Flags	Purpose	Definition / expression
Peak Cash Pool 30D	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	IF(COUNT(Daily Analytics 30 Days[Row ID]) > 0, MAX(Daily Analytics 30 Days[Outstanding Principal EOD]), 0)
Peak Cash Pool Date 30D	VC / Date	OLAP only	-	Derived date, time-window, age, or delinquency measure.	MAX(SELECT(Daily Analytics 30 Days[Snapshot Date], [Outstanding Principal EOD] = [_THISROW].[Peak Cash Pool 30D]))
Expected Daily Interest	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	SUM(SELECT(Loans[Current Daily Interest], AND([Loan Status] = "ยังไม่ปิดยอด", [Loan Type] = "ดอกเบี้ยรายวัน")))
Total Profit of Outstanding Loans	VC / Price	OLTP + OLAP	-	Derived financial aggregate or balance for operational or analytical use.	SUM(SELECT(Loans[Total Interest Received], [Loan Status] = "ยังไม่ปิดยอด"))
Profit Coverage of Outstanding Principal	VC / Percent	OLTP + OLAP	-	Derived ratio or performance measure calculated from governed source values.	[Total Profit of Outstanding Loans] / [Outstanding Principal]
Header - Outstanding Loans	VC / Text	OLTP + OLAP	-	Presentation-oriented label or bilingual display value.	"—— สินเชื่อคงค้าง / Outstanding Loans ——"
Header - Monthly Forecast	VC / Text	OLTP + OLAP	-	Presentation-oriented label or bilingual display value.	"—— พยากรณ์ / Forecast ——"
Total Cash Pool	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	SUM(Cash Pool Contributions[Net Contribution])
Total Unsettled Profit	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	SUM(Partners[Total Profit Earned]) - SUM(Partners[Completed Settlement])
Expected Loan Account Balance	VC / Price	OLAP only	-	Derived count used for workload, portfolio, or risk analysis.	[Total Cash Pool]-[Outstanding Principal]+[Total Unsettled Profit]
Header - Cash Pool	VC / Text	OLAP only	-	Presentation-oriented label or bilingual display value.	"—— เงินกองกลาง / Cash Pool ——"
Total Outstanding Principal	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	[Outstanding Principal]
Top 3 Borrower Concentration	VC / Percent	OLAP only	-	Derived AppSheet value used by the app workflow, presentation, or reporting model.	IF([Total Outstanding Principal] = 0, 0, SUM(TOP(SORT(SELECT(Borrowers[Total Outstanding Principal], [Total Outstanding Principal] > 0), TRUE), 3)) / [Total Outstanding Principal])
Borrowers Fully Paid Today	VC / Number	OLTP + OLAP	-	Derived AppSheet value used by the app workflow, presentation, or reporting model.	COUNT(SELECT(Borrowers[Row ID], [Daily Collection Status] = "Fully Paid"))
Borrowers Partially Paid Today	VC / Number	OLTP + OLAP	-	Derived AppSheet value used by the app workflow, presentation, or reporting model.	COUNT(SELECT(Borrowers[Row ID], [Daily Collection Status] = "Partially Paid"))
Borrowers with Overdue Payment	VC / Number	OLTP + OLAP	-	Derived AppSheet value used by the app workflow, presentation, or reporting model.	COUNT(SELECT(Borrowers[Row ID], [Daily Collection Status] = "Overdue"))
Borrowers Not Paid Today	VC / Number	OLTP + OLAP	-	Derived AppSheet value used by the app workflow, presentation, or reporting model.	COUNT(SELECT(Borrowers[Row ID], [Daily Collection Status] = "Not Paid"))
Header - Collection Status	VC / Text	OLTP + OLAP	-	Presentation-oriented label or bilingual display value.	"—— สถานะชำระ / Collection ——"
Amount Due Today	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	SUM(SELECT(Charges[Amount Due], [Charge Date] = TODAY()))
Amount Collected Today	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	SUM(SELECT(Charges[Total Paid], [Charge Date] = TODAY()))
Amount Outstanding Today	VC / Price	OLAP only	-	Derived financial aggregate or balance for operational or analytical use.	SUM(SELECT(Charges[Amount Remaining], [Charge Date] = TODAY()))

Column	Kind / type	Availability	Flags	Purpose	Definition / expression
Amount Collection Rate Today	VC / Number	OLAP only	-	Derived ratio or performance measure calculated from governed source values.	IF([Amount Due Today] = 0, 0, [Amount Collected Today] / [Amount Due Today])

7. Known implications and follow-up checks

Area	Implication
OLTP/OLAP variance	Shared physical tables have materially different VC layers. Changes to a shared field must be reviewed in both app dictionaries even when only one app appears to use it.
Reporting ownership	OLAP-only VCs and Reporting * slices are the preferred governed feed surface. Do not move their heavy aggregation into OLTP without a performance assessment.
Calendar source	Calendar is not in Money database. Its key/date population and refresh behavior depend on Calendar Help Table / Sheet1 and must be checked separately.
Slice comments	All inspected slice descriptive-comment fields were blank. This document supplies purposes, but future changes should also populate AppSheet descriptive comments where practical.
Delete permissions	OLTP Payment Collection currently permits updates, adds, and deletes. Any production correction to financial rows should remain auditable and avoid routine physical deletion.
Dashboard reconciliation	Reporting slice totals should be reconciled to their governed base grains and refresh cutoff before Looker Studio changes are accepted.

7.1 Manual validation checklist for the baseline

Schema: Confirm the Money database still reports 10 tables and that each physical field matches the detailed dictionary.

Relationships: Open every listed Ref Type Details panel and confirm source table plus Is a part of? state.

App variance: Confirm the OLTP and OLAP column counts shown at each table heading match this document.

Slices: Confirm row filters, included-column order, and update mode for all 13 slices.

Permissions: Verify reporting slices remain read-only and operational update access matches intended roles.

Financial logic: Sample posted, partial, fully paid, overdue, reversed/adjusted, and defaulted scenarios against the documented VC dependencies.

Reconciliation: Reconcile OLAP Reporting slice totals to Money database at a defined as-of time before dashboard use.